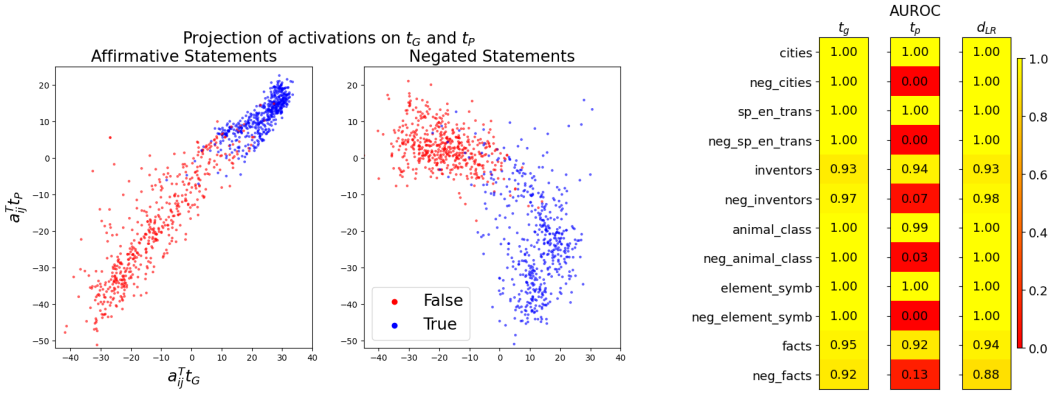

(a) (b)

Figure 11: LLaMA2-13B: Left (a): Activations $\mathbf{a}_{ij}$ projected onto $\mathbf{t}_G$ and $\mathbf{t}_P$. Right (b): Separation of true and false statements along different truth directions as measured by the AUROC, averaged over 10 training runs.

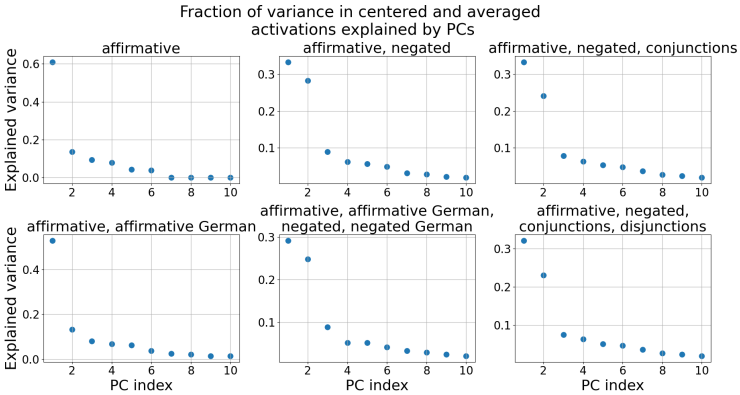

Figure 12: LLaMA2-13B: The fraction of variance in the centered and averaged activations $\tilde{\boldsymbol{\mu}}_i^+$, $\tilde{\boldsymbol{\mu}}_i^-$ explained by the Principal Components (PCs). Only the first 10 PCs are shown.

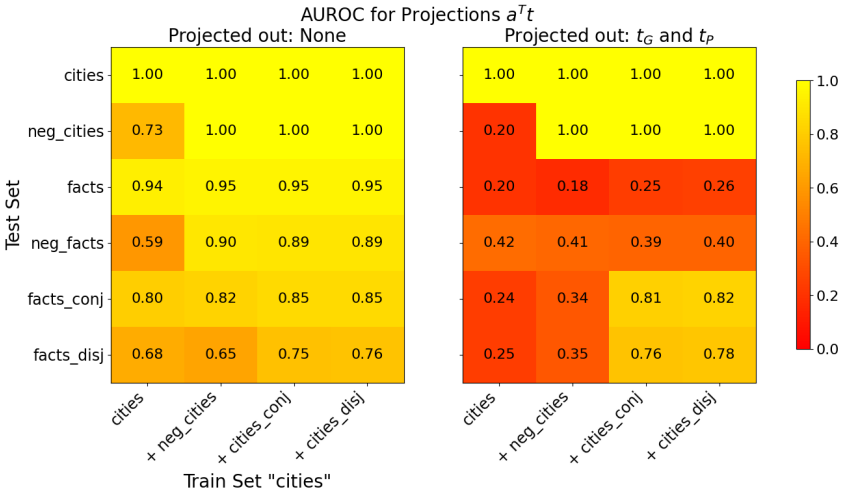

Figure 13: LLaMA2-13B: Generalisation accuracies of truth directions $\mathbf{t}$ before (left) and after (right) projecting out $\mathbf{t}_G$ and $\mathbf{t}_P$ from the training activations. The x-axis shows the train set and the y-axis the test set. All truth directions are trained on 80% of the data. If test and train set are the same, we evaluate on the held-out 20%, otherwise on the full test set. The displayed AUROC values are averaged over 10 training runs, each with a different train/test split.